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Artificial Intelligence (AI) Action Plan

Executive Summary

The rapid advancement and adoption of artificial intelligence (AI) in the United States and globally necessitate a critical examination of its current trajectory and future development. Classical AI, built on perceptron-based neural networks and powered by energy-intensive GPU/SIMD processors, has driven remarkable progress over the past decade but is now reaching its limits due to unsustainable energy consumption and escalating costs. Examples such as the $500 billion Stargate joint venture and the market disruption caused by DeepSeek's efficiency breakthrough highlight the financial and competitive pressures facing classical AI. Beyond economics, the opacity of classical AI systems raises significant concerns about transparency, trust, and potential biases, affecting individual lives and national security. Meanwhile, the concentration of AI power among a few major corporations threatens to limit access and diversity in foundational AI models, risking monopolistic control over a technology critical to the nation's future.

To address these challenges, the United States must invest in both sustaining classical AI and pioneering its successor, such as neuromorphic computing, which promises greater efficiency, reduced energy use, and transparent decision-making. Additionally, common-sense regulations are needed to ensure AI transparency, protect against discrimination, and democratize access to AI technologies. By learning from precedents like the EU's GDPR and historical interventions such as the AT&T breakup, the U.S. can maintain its leadership in AI innovation while safeguarding its citizens and national interests.

Innovation in non-classical AI:

The AI industry is currently dominated by classical AI. To be clear, classical AI refers to systems that are centered around perceptron-based neural networks (the model) and are accelerated by GPU/SIMD type processors. While it is undeniable that this classical AI has made incredible advances over the past decade, it is also true the scale we are currently operating at is pushing classical AI to a breaking point. Chief amongst the problems posed by classical AI is the tremendous energy consumption of these systems. In fact, two recent events underscore the problem we face with classical AI systems…

1. The announcement of the Stargate joint venture. While investment in AI infrastructure is in the national interest, the announced $500B expenditure by Stargate is primarily driven by the massive scale of the data centers needed to house classical AI systems. The huge energy requirements and volumetric space required to accommodate classical AI is clearly evident in the scale of investment required to house these systems. Adjusted for inflation, the cost of this AI infrastructure is 10X higher than the construction costs of the Hoover Dam.

2. A relatively obscure Chinese AI company, DeepSeek, announced a breakthrough in AI efficiency which resulted in a staggering $1T loss in the US/Euro tech markets. This is a clear indicator that AI efficiency is front and center on the minds of the entire AI industry. The exorbitant cost of classical AI is already painful today, and it will only continue to get worse as our foundational models continue to grow in size and AI continues on it's blistering pace to embed itself in American society.

Although the two events discussed above discuss the financial costs of classical AI, there is another ‘cost’ to classical AI that is just as concerning. Considering that the concepts associated with classical AI are more than 50 years old, and that the entire industry is compelled to discover a new approach to AI, it follows that there is a competition amongst developed nations to discover what comes next. Just as the steam engine gave way to the internal combustion engine, we can be certain that somewhere, some group of researchers, private entity or government will break through the barriers posed by classical AI. Whoever achieves this will harvest not only an economic benefit, but quite likely a military benefit as well.

Consider Intel Corporation and the technological advantage that was afforded the United States by leading the world in the microprocessor revolution. Those benefits accrued for decades and are only now starting to diffuse. For AI, the stakes will be even larger. At this moment it is clear that China is pushing hard in AI R&D. They have already surpassed the US in the important metric of producing published research focused on AI. In fact, in 2017 China announced their ‘New Generation Artificial Intelligence Development Plan’ in which they seek to lead all nations in AI by 2030. They are well on their way.

In order for the US to maintain AI superiority, we must invest in both classical AI systems as well as funding the research for what will inevitably replace classical AI systems. One of the more promising emerging technologies is referred to as ‘nueromorphic computing’. Neuromorphic computing systems leverage the most contemporary neuroscience to fully unlock the superior processing and intelligence of the human brain. The benefits include massive reductions in energy use and implementation costs. Additional benefits include full model transparency so decisions so that human operators and government regulators can have higher confidence in the use of AI throughout American industry.

There should be no doubt that the classical AI systems that are being adopted across every corner of American industry will give way to something more efficient and even more powerful. The US must proactively make the investments necessary to ensure we are the nation to discover the next engine that will power the AI industry.

The need for AI Transparency:

AI is continuing to be rapidly adopted by nearly every industry in the US and worldwide. Consequently, more and more decisions that impact the lives of citizens throughout the world is being relegated to AI systems that are hidden and out of view of the public. The impact of decisions made by these classical AI systems can be acutely personal, as in the approval or denial of a loan application. Consider too, the use of AI to develop a treatment plan for a life-threatening disease. It is understandable that both patient and doctor would want the very highest level of confidence in this type of AI system. Beyond the impact to individuals, the decisions made by AI systems can also have implications to the national security of the United States. For example, imagine an AI system controlling a fire control system for US Air Defense. At the critical moment of designating an unknown aircraft as 'friend' or 'foe' it is vitally important to have confidence in the AI system that is making that determination and the basis for its recommendation. The wrong decision in this type of scenario can have catastrophic consequences.

Unfortunately, the underlying mathematics of classical AI systems make it very difficult, perhaps impossible to clearly understand the basis for any recommendation that comes from AI systems. Indeed, we have already seen documented examples of unintended bias creeping into AI systems that can cause damage to individual citizens. The resulting lack of trust in AI threatens to limit the further adoption of AI by certain industries, in spite of the efficiencies that can be realized through the adoption of AI.

Beyond the individual, lack of transparency in AI systems can lead to large scale bias against entire groups of our population. In other words, opacity in AI systems can lead to widespread discrimination that is very hard to detect. The remedy to all of these problems is to provide common sense regulation of AI that is used in applications that can impact individuals and our national security. The consequences of an AI system making the wrong choice while sorting freshly harvested produce is one thing, but the consequence of an AI system discriminating against a class of individuals is quite another. The US has created laws that prohibit discrimination, but what happens when that discrimination is deeply embedded in a system that has no way of being inspected?

The European Union has already implemented important safeguards from AI systems for its citizens. The 'Right to an Explanation' clause of GPDR (General Data Protection Regulation) stipulates that individuals have a right not to be subject to a decision based solely on automated processing. According to GPDR, if automated decision processing (AI) is used, affected individuals have the right to obtain meaningful information about the logic involved. Unfortunately, the decision logic in classical AI systems is difficult to ascertain. This leads to businesses avoiding the use of AI, not because it provides the wrong decisions but rather because the decisions it provides cannot be explained.

Until we move beyond classical AI and the opacity that surrounds it's decisions, common sense regulations must be put in place to protect individuals. In addition to appropriate regulations, investments in the development of AI that exhibits full transparency in the decision process should be undertaken.

Democratization of AI:

AI systems are growing at an unprecedented scale. As evidenced by the recent Stargate announcement, investments of $500B will be made to house the massive AI systems that are being designed for near-term use. The massive capital expenditures to power AI aren't the only problem. These massive AI systems require equally massive amounts of information to train the AI systems. The sheer size of the capital investment and amount of data needed to train these systems presents a significant barrier to entry in the AI market, especially at the highest end where foundational AI models rule the day. Already today we see the emergence of a highly concentrated group of companies that can afford to operate at this scale. This group includes corporations like Google, Microsoft and Meta and a few others in the US and abroad. As we go forward, it is possible for this group of AI 'power players' to further consolidate as the market shakes out and the most successful models begin to achieve market dominance.

The US has long recognized and regulated the existence of monopolistic enterprises. This is especially true when those enterprises control critical commodities or infrastructure that are of national importance. This same approach must be taken with AI. Equal access and diversity of choices in foundational AI models must be part of a plan for the US going forward. Furthermore, both users of these systems and the government officials who will regulate these systems should have access to decision logic or the underlying construction of the models. This is particularly true for any models that use post processing 'tune' decisions towards some perceived ideal.

The break-up of AT&T serves as an example of how the US Government has intervened to protect the delivery of critical services to the citizens of the US. In the case of AI, the stakes may be even higher. Imagine the potential implications if our most frequently used 'source of truth' has deeply embedded ulterior motives. Already we see the potential for biased media to impact matters of national significance, like US elections. Biases in AI systems, whether intentional or not, can have long reaching impacts on the US. Here again, common sense regulations that work to democratize (providing access to, choice of and insight to) the emerging realm of AI will be in the best interest of the US and governments of developed nations.

Call to Action

The US can create opportunities to lead the world in cutting edge AI based on neuromorphic computing by taking these three steps.

- Fund Research and Development: Allocate significant federal funding to accelerate research into neuromorphic computing, supporting universities, national labs, and private-sector partnerships to unlock its potential as the next-generation AI technology.
- Incentivize Adoption: Establish grants, tax credits, or public-private initiatives to encourage industries to transition from energy-intensive classical AI to neuromorphic systems, reducing costs and environmental impact.
- Move Neuromorphic AI to The Edge: The American war fighter deserves the best technology available to achieve their mission goals. Neuromorphic computing provides unique capabilities in terms of SWAP (Size Weight and Power), explainability, and functionality. Invest to projects to bring neuromorphic AI alternatives to the battlefield.

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